RAG-NoiseLaw??A Retrieval-Augmented Framework for Noise Regulation Question

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Abstract

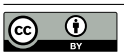
As urbanization accelerates, environmental noise pollution poses increasing risks to public health and well-being. However, existing regulatory frameworks are often fragmented and complex, making it difficult for both the public and enforcement agencies to retrieve and interpret relevant information. To address this, this study introduces a retrieval-augmented question-answering system based on the RAG framework, integrating a large language model with a domain-specific knowledge base. The system enhances factual accuracy and contextual relevance by combining semantic retrieval over legal and environmental data with answer generation. Empirical evaluations confirm significant improvements in both retrieval efficiency and response quality, offering a practical tool for interpreting noise regulations.

CCS Concepts

• **Information systems**; • **Information retrieval**; • **Retrieval tasks and goal**; • **Question answering**;

Keywords

Retrieval-Augmented Generation, LLMs, Knowledge Retrieval, Urban Noise Regulation



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1 Introduction

With the accelerated pace of urbanization and industrial development, environmental noise pollution has emerged as a pervasive and escalating threat to public health, quality of life, and ecological sustainability [5]. Chronic exposure to noise has been linked to a range of adverse health outcomes, including sleep disturbances, cardiovascular issues, and cognitive impairments [6]. In response, many countries have promulgated noise control legislation and developed environmental monitoring infrastructures. However, despite these institutional efforts, the practical utility of regulatory documents and monitoring data remains limited in real-world applications. Legal texts are often lengthy, jurisdiction-specific, and written in highly formalized language, while environmental monitoring data is fragmented, inconsistently formatted, and typically stored in unstructured reports. This heterogeneity impedes the efficient retrieval, interpretation, and use of critical information by both the public and regulatory professionals [1].

In parallel, Large Language Models (LLMs) have demonstrated state-of-the-art performance across a wide range of general-purpose NLP tasks, showing strong capabilities in text comprehension, reasoning, and generation [11]. However, their applicability in specialized domains—such as legal compliance and environmental regulation—remains constrained. Traditional LLMs store knowledge

in their parameters, which limits their ability to capture dynamically evolving, localized, and domain-specific information [10]. As a result, they are prone to producing hallucinations, factual inconsistencies, and temporally outdated responses when applied to legal and environmental contexts [7]. Such limitations reduce their reliability in tasks where interpretability, accuracy, and verifiability are paramount.

To overcome these limitations, this work presents a Retrieval-Augmented Generation (RAG)-based question answering (QA) framework specifically tailored for the domain of noise regulation [8, 15, 25]. The proposed system integrates heterogeneous knowledge sources, including national and regional legal provisions, environmental standards, and unstructured noise monitoring reports—into a unified framework that supports accurate and context-aware QA. Unlike conventional end-to-end generative models, which often generate responses based solely on internal parameters, our system first retrieves relevant documents via semantic search and then generates grounded responses with explicit source citations. This hybrid architecture improves the factual correctness, domain fidelity, and explainability of the answers.

To cope with the structural disparity between formal legal language and semi-structured or unstructured environmental data, specialized preprocessing pipelines were designed. Legal documents are segmented using fixed-length chunking methods suitable for hierarchical text, while environmental reports are parsed using semantic-aware chunking to preserve contextual coherence. This facilitates effective matching between user queries and relevant textual evidence during retrieval.

In general, our work has the following contributions:

(1) Proposed RAG-Based Regulatory QA System for Environmental Noise We propose a Retrieval-Augmented Generation (RAG) question answering system specifically tailored for the domain of regulatory noise governance, leveraging the capabilities of large language models. The system demonstrates several distinctive advantages: ① Heterogeneous Data Integration: Legal texts and environmental monitoring data are systematically integrated into a unified, semantically indexed, and retrievable knowledge base, enabling cross-format information access; ② Domain-Specific Terminology Representation: Specialized legal and technical terms are accurately embedded within the vector space, enhancing the semantic alignment between queries and content in a domain-sensitive manner; ③ Efficient and Accurate Retrieval: The retrieval component is optimized to balance semantic search precision with computational efficiency, ensuring responsive performance under real-time constraints; ④ Stable and Trustworthy Generation: The generative model produces factually grounded, readable, and traceable answers, with particular robustness in multi-hop reasoning and cross-modal inference scenarios.

(2) Construction of a Domain-Specific Benchmark and Evaluation To empirically validate the proposed approach, we constructed a domain-specific benchmark dataset encompassing a wide range of regulatory and environmental scenarios related to noise management. The dataset captures diverse user intents and governance contexts. Experimental evaluations demonstrate that our system significantly outperforms baseline models in both retrieval accuracy and answer quality. These results underscore the system's potential to support public engagement, compliance auditing, and data-informed decision-making in environmental governance through a scalable and reliable AI-powered framework. To validate our framework, a domain-specific benchmark were established dataset encompassing a variety of noise-related regulatory and environmental scenarios. Empirical evaluations comfire that our approach significantly outperforms baseline models in both retrieval accuracy and answer quality, offering a scalable and trustworthy solution for supporting public engagement, compliance checks, and decision-making in environmental governance.

The structure of this paper is organized as follows. **Section 2** provides a comprehensive review of related work on Retrieval-Augmented Generation (RAG), highlighting its key methodologies and applications in relevant domains. **Section 3** details the preprocessing procedures applied to noisy data and presents the overall architecture of the RAG framework adopted in this study. **Section 4** discusses the experimental design, including parameter configurations, model selection strategies, and evaluation metrics used to assess model performance. Finally, **Section 5** concludes the paper by summarizing the main findings and proposing potential directions for future research and optimization.

2 Related Work

Open-domain question answering (ODQA) has witnessed substantial progress with the advent of retrieval-augmented architectures, which integrate external knowledge to enhance the factual accuracy and contextual relevance of responses. Early QA systems primarily relied on the Probabilistic Relevance Framework (PRF), such as BM25, which uses lexical overlap and TF-IDF weighting to rank candidate documents [19]. While effective in general-purpose retrieval, these methods lack semantic understanding, limiting their utility in complex, domain-specific tasks—particularly in areas such as legal interpretation and environmental policy, where precision and contextual depth are critical [13].

To address these limitations, dense retrieval methods have emerged, with Dense Passage Retrieval (DPR) introducing semantic embeddings learned via dual encoders, replacing sparse vector representations [12]. This allows for more nuanced query-document matching by projecting both into a shared semantic space and capturing deeper contextual signals beyond token-level similarity.

Building on this paradigm, the Retrieval-Augmented Generation (RAG) framework integrates dense retrievers with large-scale generative language models. By conditioning generation on retrieved evidence, RAG significantly improves the factual grounding of outputs in knowledge-intensive settings [16]. It has proven effective in reducing hallucinations in dialogue systems [20], and benefits from modular improvements such as semantic chunking [14], cross-encoder re-ranking [17], and masked autoencoder-based query rewriting [18]. Recent advancements include hybrid retrievers, dynamic embeddings, and reinforcement-trained rewriters to further improve retrieval precision and robustness [9, 21].

To tackle the unique challenges of noise regulation—marked by fragmented legal provisions and heterogeneous environmental monitoring data—we propose a domain-adapted question answering (QA) system grounded in the RAG framework. Our method

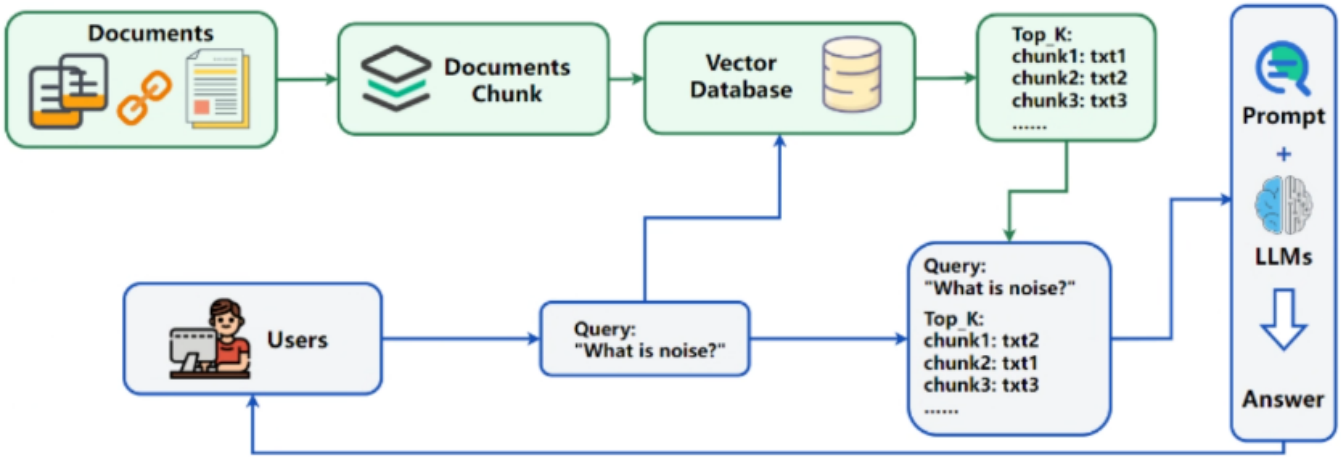


Figure 1: RAG-NoiseLaw architecture

incorporates a unified preprocessing pipeline including text normalization, semantic chunking, and metadata alignment to bridge structural disparities across data sources . A dense retriever encodes both user queries and textual chunks into a shared vector space, retrieving top-ranked segments via Maximum Inner Product Search (MIPS). These are passed to a generator that produces grounded, context-aware responses [20].

To further enhance domain adaptability, we incorporate query rewriting techniques and adopt advanced foundation models such as LLaMA 2 as the backbone generator [21]. Our design draws from cross-domain adaptations of RAG, including MEDRAG in biomedical QA [24], Lawformer in legal applications [23], and IIKM in industrial knowledge systems [4], highlighting the framework's extensibility across specialized domains.

By adapting RAG to environmental noise governance, our system bridges the semantic gap between regulatory norms and observational data, enabling accurate, traceable, and semantically robust QA. This work also lays the foundation for future research in multi-hop reasoning, cross-format embedding alignment, and low-latency, real-time regulatory response systems.

3 Methodology

3.1 Data preprocessing

To construct a high-quality noise domain knowledge base, we systematically processed heterogeneous data sources, primarily consisting of legal and regulatory documents and environmental noise monitoring reports.

Legal and regulatory documents are often published in complexly formatted PDFs, which pose challenges for direct processing due to issues such as structural disorder and fragmented semantics. To improve readability and parsing, we converted these documents into Word format and utilized OCR and text extraction tools to extract the main content. Subsequently, we performed paragraph segmentation, noise-related topic labeling, and formatting standardization to facilitate semantic segmentation and embedding processes in later stages.

The monitoring report data primarily originates from Excel spreadsheets provided by enterprises or regulatory bodies. Initial attempts to input this structured data directly into the RAG framework revealed significant shortcomings, as the lack of semantic continuity in structured tables led to poor retrieval performance. This issue became more pronounced as the data scale increased.

To optimize this process, we explored two improvement strategies: ①**Transformation to Markdown Format:** We converted the table contents into a more readable Markdown structure, which facilitated the model's ability to recognize the relationships between table headers and data, thereby improving retrieval accuracy in certain scenarios; ②**Semantic Conversion to Natural Language Descriptions:** We further converted key fields and data in the tables into natural language paragraphs. For example, the sentence "The daytime compliance rate for Monitoring Point A is 90%, meeting the monitoring standard" was treated as an atomic information unit. After semantic segmentation, the text was processed into paragraphs and passed to the vectorization module for index construction.

Empirical evaluations confirmed that transforming structured data into natural language text and applying semantic segmentation are critical methods for enhancing retrieval performance in the RAG system. These findings underscores the importance of semantic unification and preprocessing of heterogeneous data sources prior to embedding, providing a reusable paradigm for future large-scale knowledge base construction.

3.2 Overrall architecture

Retrieval-Augmented Generation (RAG) combines neural retrieval mechanisms with generative pre-trained models (e.g., BERT, GPT, BART). It aims to address the limitations of parameterized knowledge in LLMs by enhancing factual accuracy and knowledge controllability. By integrating scalable external knowledge sources, RAG implements a "retrieval-generation" [22] dual mechanism, effectively mitigating hallucination [20] issues.

In this study, the RAG framework is applied to develop an intelligent question-answering system for urban noise management. The

core process, as depicted in Figure 1, includes three stages: ①**Query Reception:** The user submits a noise-related query, such as "What is the maximum allowable noise level during the day for industrial enterprises?"; ②**Semantic Retrieval:** The system performs vectorized retrieval from the constructed noise knowledge base to select the most relevant text passages; ③**Answer Generation:** The query and retrieved passages are fed into a generative language model, which synthesizes a fact-based, natural language response, e.g., "According to the 'Industrial Enterprise Boundary Noise Emission Standard,' the daytime limit is 65dB."

This RAG framework generates answers by combining user queries with vectorized knowledge base segments, enhancing the model's ability to understand legal expressions and improving its robustness in multi-source, heterogeneous knowledge environments.

4 Experiment

To construct a high-quality question-answer (QA) dataset for noise regulation, a hybrid methodology was employed that combines domain expert feedback with large language model (LLM) generation. Real-world questions were first collected from regulatory personnel, then expanded using LLMs to enhance coverage and linguistic diversity. As shown in Figure 2, these questions are processed within a RAG-based pipeline. Each query is encoded and matched to a structured knowledge base—comprising regulatory documents, standards, and reports—using a dense retriever. The top-K relevant passages are retrieved and passed to a generative model, which produces grounded answers along with source citations. This workflow facilitates high domain relevance, semantic diversity, and factual grounding. It enhances the credibility of responses while supporting traceability and regulatory transparency.

Table 1: Hyperparameter Setting.

	Parameter	Value
Chunking	Chunk Size	512
	Overlap Size	50
LLMs	temperature	0.1

In this experiment, a unified preprocessing and construction process was used, including text chunking, information retrieval, and answer generation. Legal and regulatory documents were chunked using a fixed-length mechanism, while report data was segmented to maintain semantic coherence. Top_K relevant passages were retrieved in each query, and answers were generated using the model. All experiments were conducted using a consistent test set and evaluation metrics for comparison. The hyperparameter settings are listed in **Table 1**, which include a chunk size of 512, an overlap size of 50, and a temperature parameter of 0.1 for the LLM.

To rigorously assess the effectiveness and robustness of the proposed RAG-based domain-specific QA system, a comprehensive evaluation protocol was designed encompassing both the retrieval and generation components. A curated test set comprising 15 representative question types was constructed, each reflecting a distinct category of user intent or regulatory inquiry scenario relevant to the domain of environmental noise governance. These question types were selected to ensure broad coverage across application contexts, including factual queries, procedural clarifications, threshold-based assessments, legal reference identification, and multi-source data integration challenges.

Table 2: Information Retrival Effect Evaluation

Embedding Model	MRR	Recall@k	MAP
BGE-M3	100.00%	100.00%	72.22%
BGE-Large	70.56%	80.00%	42.76%
E5-Large	90.00%	93.33%	58.33%
Snowflake	74.67%	85.22%	41.33%

In **Table 2**, we conducted a comparative analysis across four state-of-the-art sentence embedding models for the retrieval module: BGE-M3 [2], BGE-Large, E5-Large, and Snowflake. Each embedding model was used to encode the document corpus and queries into a high-dimensional vector space. Relevance was computed using Maximum Inner Product Search (MIPS), and the top-k retrieved documents were analyzed using standard information retrieval metrics: ①**Mean Reciprocal Rank (MRR)**, which evaluates the rank position of the first relevant document; ②**Recall@k**, measuring the proportion of relevant documents retrieved within the top-k candidates; ③**Mean Average Precision (MAP)** [3], which assesses both precision and recall across ranked lists.

This evaluation aimed to determine the semantic alignment fidelity of each embedding model under domain-specific constraints, particularly in handling lengthy, fragmented legal texts and numerically dense environmental records.

For the generative component, the performance was assessed of four large language models (LLMs): ChatGLM, Qwen-Max, DeepSeek-Chat, and Baichuan4. Each LLM was conditioned on the top-k retrieved documents from the retrieval stage and tasked with generating final answers to the test queries. The generated outputs were manually and automatically assessed based on the following dimensions: ①**Accuracy**: Whether the response correctly reflects the authoritative content of the retrieved documents; ②**Completeness**: The extent to which the response addresses all relevant aspects of the question; ③**Hallucination Rate**: The proportion of generated content not supported by retrieved evidence, critical in legal and policy-sensitive applications; ④**Readability**: Evaluated using both human ratings and linguistic coherence scores to measure the fluency and clarity of the responses. Each model's performance was further analyzed by cross-referencing generation quality with retrieval fidelity to understand the interaction between upstream document relevance and downstream answer generation quality.

In summary, the proposed evaluation protocol offered a rigorous and multidimensional benchmark for assessing both the retrieval and generation modules of the RAG-based domain-specific QA system, specifically tailored to the regulatory context of environmental noise governance. By designing a test suite of 15 diverse question types—ranging from statutory interpretation to data integration—the study ensured comprehensive scenario coverage.

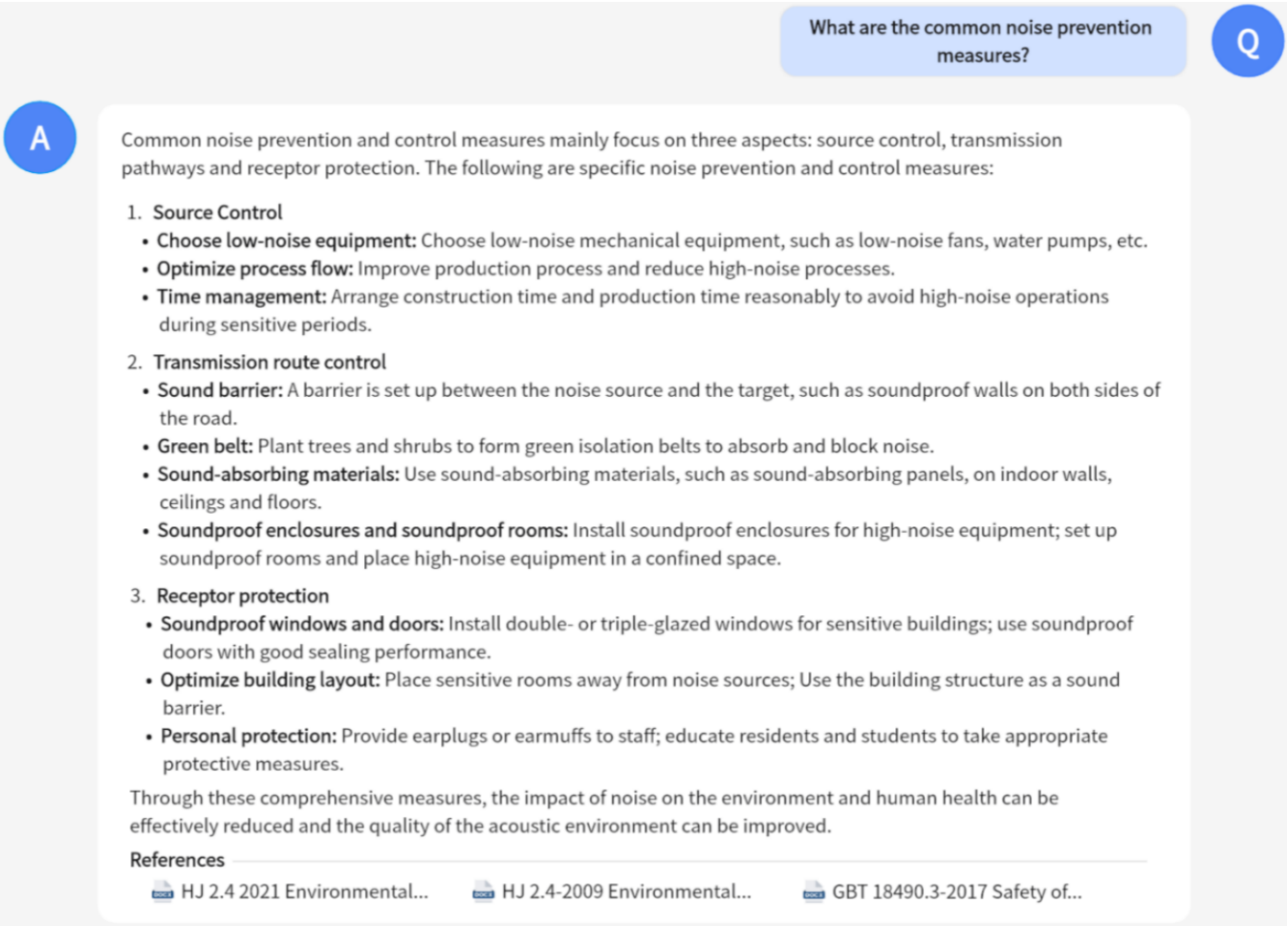


Figure 2: Question and answer process

Table 3: Answer Generation Quality Assessment

LLMs	Accuracy	Completeness	Hallucination Rate	Readability
ChatGLM	86.67%	76.67%	0.67%	86.00%
Qwen-Max	92.67%	83.33%	0.33%	93.33%
Deepseek-Chat	94.67%	90.67%	4.67%	96.00%
Baichuan4	90.00%	75.33%	0.33%	72.00%

The retrieval component was analyzed across four types of embedding models using MRR, Recall@k, and MAP, effectively quantifying semantic alignment. Concurrently, the generative performance of four large language models was evaluated based on accuracy, completeness, hallucination rate, and readability, ensuring factual grounding and linguistic coherence.

The results underscore the crucial impact of embedding selection on downstream answer quality, highlighting the importance of tightly integrated retrieval-generation pipelines in domain-adaptive QA. This study provides a solid foundation for future research and real-world applications in knowledge-intensive, regulation-driven environments.

BGE-M3 demonstrated the best performance in information retrieval, achieving 100% in both MRR and Recall@k, indicating its superiority in retrieval accuracy and comprehensiveness. E5-Large followed closely, with slightly lower MRR and Recall@k scores but a stronger MAP performance. **Table 3** presents a comparative evaluation of four large language models (LLMs) on answer generation quality across four criteria: accuracy, completeness, hallucination rate, and readability. Among them, DeepSeek-Chat achieves the

highest accuracy (94.67%) and completeness (90.6%), while Qwen-Max demonstrates strong overall performance with superior readability (93.33%) and the lowest hallucination rate (0.33%). ChatGLM and Baichuan4 exhibit moderate performance, with ChatGLM offering balanced results and Baichuan4 showing reduced readability (72.00%) despite maintaining competitive accuracy. These findings indicate that both DeepSeek-Chat and Qwen-Max possess strong generation capabilities and are well-suited for complex regulatory question answering tasks where both fluency and factual reliability are essential.

5 Conclusion

In response to the challenges posed by the heterogeneity and unstructured nature of legal provisions and environmental noise monitoring data, this paper proposes a retrieval-augmented generation (RAG) based question answering (QA) framework tailored for noise regulation domains. We design a unified data preprocessing pipeline to standardize multi-source inputs and enable semantic-level alignment between regulatory texts and monitoring reports. Our approach integrates adaptive embedding models with large language models (LLMs) to perform dense retrieval via maximum inner product search and to generate contextually grounded, regulation-aware answers. Empirical evaluations confirm notable improvements in both retrieval precision and response quality, particularly in mitigating factual hallucinations commonly observed in LLMs when applied to domain-specific queries. This work provides a practical pathway toward robust QA systems in regulatory environments. Future work will explore multimodal knowledge alignment, dialog-based interaction, and continuous knowledge update mechanisms to support evolving real-world noise governance needs.

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